

Department of the Interior
U.S. Geological Survey

**Landsat 10 (L10)
Design Reference Case – 18 Days (DRC-18)
Definition**

Landsat 10

Version 2.0

May 2026



Landsat 10 (L10)

Design Reference Case – 18 Days (DRC-18) Definition

May 2026

EROS
Sioux Falls, South Dakota

Executive Summary

The Landsat 10 Design Reference Case – 18 Days (DRC-18) provides an overview of the mission operational image acquisition scenarios over an 18-day period.

Landsat represents the world's longest continuously acquired collection of space-based moderate-resolution land remote-sensing data. The program's imagery provides a unique resource for those who work in agriculture, geology, forestry, regional planning, education, mapping, and global change research. Landsat images are also invaluable for emergency response and disaster relief.

This document is under the U.S. Geological Survey (USGS) Landsat 10 (L10) Project Configuration Control Board (CCB) control. Please submit changes to this document, as well as supportive material justifying the proposed changes, via a Change Request (CR) to the Configuration Management (CM) tracking tool.

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Section 1 Introduction

1.1 Background

The Landsat 10 (L10) mission is a component of the Sustainable Land Imaging (SLI) Program conducted jointly by the National Aeronautics and Space Administration (NASA) and U.S. Geological Survey (USGS) of the Department of the Interior (DOI).

L10 was conceived as a follow-on mission to the highly successful Landsat series of missions that have provided satellite coverage of Earth's continental surfaces since the launch of the Earth Resources Technology Satellite (ERTS, later renamed Landsat 1) in 1972. The Landsat data archive constitutes the longest continuous moderate-resolution record of the global land surface as viewed from space. L10 is intended to ensure that Landsat-like data be provided to the USGS National Satellite Land Remote Sensing Data Archive (NSLRSDA) for a period of not less than five years.

1.2 Purpose

This document provides an overview of the L10 mission image acquisition scenarios over an eighteen-day period referred to as the Design Reference Case – 18 Days (DRC-18). The scenario spans 18 days because that is the repeat period of the mission Worldwide Reference System-3 (WRS-3) grid.

The DRC-18 itself is stored in an Excel spreadsheet and is tracked with this document as one Configuration Item (CI) via Document Control Number (DCN) LSDS-2389. It is available on the Landsat 10 Configuration Management System.

The DRC-18 currently documents the image acquisitions for the L10 observatory over the 18-day WRS-3 repeat cycle.

1.3 Scope

This document describes the image acquisition cadence for the L10 mission. All references to instrument calibration activities are based upon the current operations concept known at this time. Specific instrument calibrations for the mission will be determined by the L10 instrument provider and the joint USGS and NASA Calibration and Validation (Cal/Val) Team with the results being incorporated into follow-up versions of this document as they are known.

1.4 References

1.4.1 Applicable Documents

The documents in Table 1-1 are considered to form a part of this document.

Document Number	Document Title
LSDS-2389	Landsat 10 DRC-18 Spreadsheet
L10-SYS-DESC-0018	Landsat 10 Worldwide Reference System-3 (WRS-3)

Table 1-1. Applicable Documents

1.4.2 Reference Documents

The documents in Table 1-2 provide additional background and context for this document.

Document Number	Document Title
L10-MGMT-REQ-0001	Landsat 10 Program-Level Requirements Appendix (PLRA)
L10-SYS-REQ-0002	Landsat 10 Science and Mission Requirements Document (SMRD)
L10-SYS-PLAN-0007	Landsat 10 Mission Operations Concept (Ops Con) Document
LandIS-SE14-001	LandIS Instrument Operations Concept Document (SE-14)

Table 1-2. Reference Documents

Section 2 Landsat 10 Overview

2.1 Mission Statement

L10 will continue the acquisition, archive, and distribution of multi-spectral imagery affording global, synoptic, and repetitive coverage of Earth's land surfaces and coastal regions at a scale where natural and human-induced changes can be detected, differentiated, characterized, and monitored over time.

2.2 Summary

The L10 mission is in keeping with the Landsat programmatic goals stated in the Commercial Space Act of 1998 (Public Law 105-303) and the Land Remote Sensing Policy Act of 1992 (Public Law 102-555), which require the Landsat program to provide future data that are consistent with previous Landsat data, allowing the detection and quantitative characterization of changes in or on Earth's land surfaces.

L10 continues the long-running partnership of NASA and USGS, with NASA providing the Space and Launch segments and USGS providing the Ground System and Mission Operations. NASA/Goddard Space Flight Center (GSFC) will provide overall L10 project management, mission system engineering, and mission assurance during development and will transition the mission to USGS for science operations following on-orbit commissioning.

The L10 mission features a single observatory operating in a 653 kilometer (km) sun-synchronous, low-Earth orbit. The observatory includes the Landsat Instrument Suite (LandIS), an integrated set of sensors designed to capture the full imaging swath needed to achieve global coverage on a repeating 18-day ground track. The instrument will provide 26 spectral bands with a maximum Ground Sampling Distance (GSD) of 10-60 meters (m) dependent on the spectral band, covering a swath of approximately 164 km on a ground track defined by a Worldwide Reference System known as WRS-3.

2.3 Objectives

L10 has these major mission objectives:

- Collect and archive moderate resolution Visible to Shortwave-Infrared (VSWIR) reflective and Thermal Infrared (TIR) emissive image data affording seasonal coverage of the global landmass for a continuous period of not less than five (5) years.
- Ensure that L10 data are sufficiently consistent with data from the earlier Landsat missions in terms of relative acquisition geometry, calibration, coverage characteristics, spectral characteristics, output product quality, and data availability; all of which permit scientific studies, management of land cover and land use change, along with associated inland and coastal water resources, over multi-decadal periods.
- Ensure L10 is responsive to documented emerging user needs and applications as characterized by periodic assessment, currently the User Needs for the Sustainable Land Imaging Program July 2018, Release 2.0,

and identified by the operational requirements for acquisition, processing, archiving, and distribution of Earth's image data to the U.S. Government and other users.

- Distribute L10 data products to the public freely and on a nondiscriminatory basis.

For detailed information on the L10 mission overview and architecture, and operations concepts, reference the L10-SYS-PLAN-0007 Landsat 10 Mission Operations Concept (Ops Con) Document.

Section 3 Landsat 10 Design Reference Case Overview

3.1 Key Features

There are several key features for the activities that take place in the 18-day period depicted in the DRC-18. They are:

- The DRC-18 starts at path 122, row 1 on the WRS-3 grid, which is near the North Pole
- The start date for the DRC-18 is March 12, 2022. There is no contractual or requirement implication to this start date. With this start date it is possible to perform a monthly lunar calibration maneuver.
- Each day has approximately 14.8 orbits
- The path/row pairs represent coordinates in the WRS-3 grid
- There are both imaging and instrument calibration activities

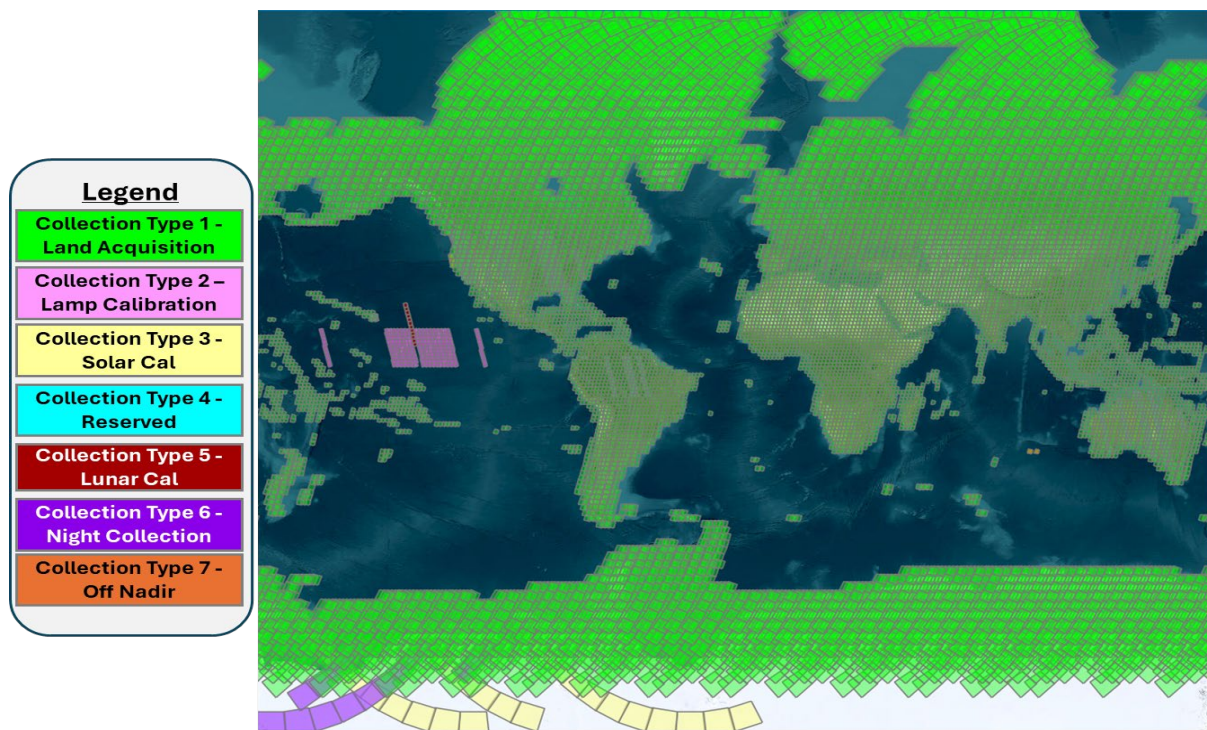


Figure 3-1. DRC-18 All Activities

3.2 Spreadsheet Format

The DRC-18 is provided as a Microsoft Excel spreadsheet and is tracked with this document as one Configuration Item (CI) via Document Control Number (DCN) LSDS-2389. There is one row containing the names of all the columns, followed by 65,720 rows of data. There are 265 paths in the WRS-3, and 248 rows per path, resulting in 65,720 distinct path-row locations. The detailed definition for each column in the DRC is provided in Table 3-1.

Column Name	Description
DRC Day	This field counts “UTC Days” as covered by the DRC, with numbering starting at 1. It does NOT count whole 24 hour periods starting from the first second of the DRC.
Scene Center Time	The Coordinated Universal Time (UTC) of the center point for each WRS-3 Path and Row location in the table
Path	The WRS-3 Path number
Row	The WRS-3 Row number
Collection	A single value denoting what science operation takes place in this location, consistent with the following mapping: 0 – no activity 1 – Nominal Earth Imaging Acquisition including Blackbody and Space View calibration collections on each scan 2 – Lamp Calibration 3 – Solar Calibration 4 – Reserved 5 – Lunar Calibration 6 – Night Imaging or Collection with Sun Angle < 5 degrees 7 – Off-Nadir Acquisition
Sun Angle	The Solar incidence angle to the WRS-3 scene center at time of collection. This value does not consider terrain features. The scene is illuminated when values are positive and in the dark when the values are negative.

Table 3-1. DRC-18 Table Definitions

The entries in the DRC-18 table are arranged chronologically, in the order flown. This time-ordered sequence of high-level operations is provided to allow analysis of the impacts and supportability of these operations, whether by spreadsheet or more elaborate means.

3.3 DRC-18 Generation Process

With the generation of the new WRS-3 definition for L10, a new set of acquisitions is needed. To generate the land imaging acquisitions, the WRS-3 grid was overlayed onto the WRS-2 grid to provide a mapping between which WRS-3 scenes overlapped which WRS-2 scenes. The Landsat 9 (L9) Land Collection Request (LCR) is used by the Landsat Flight Operations Team (FOT) to determine which WRS-2 scenes should be routinely collected by the Landsat 8 (L8) and L9 satellites. WRS-3 scenes were identified as potentially being collected if any part of the scene overlapped a WRS-2

scene that is contained within the current LCR. These potential scenes are then evaluated with the criteria outlined in the following sections to determine if they should be acquired. The end result is that the DRC-18 captures at least all of the areas currently covered by L9 LCR operations. It is expected that further refinement of the L10 acquisitions will occur in coordination with the Science Team and will be reflected in future updates of this document.

3.4 Narrative Description

This section presents a narrative overview of the DRC-18. The operations referenced in this section are described in more detail in the Mission Operations Concept Document (L10-SYS-PLAN-0007).

3.4.1 Overview

This 18-day period of operations reflects the L10 acquisition pace for the given time of year (March). As the duration of sunlight varies throughout the year, the distribution of imaging also varies.

There are three primary types of science operations described in this DRC: nadir-looking sunlit land; selective night imaging, and a variety of calibrations. Each is described in the following sections.

3.4.2 Sun-lit Land (Collection Type-1)

The core science component of the DRC-18 is the nadir-viewing Earth imaging data collections, also referred to as “nominal Earth imaging.” These are scheduled over all land areas that are illuminated by sunlight at an angle of 5 degrees above horizontal, or greater. The Landsat project maintains a land database that defines all the targets of interest, which also include coastal/littoral targets, coral reefs, small islands and some land-fast ice/glaciers.

The amount of sun-lit land being flown over varies across the 18 days of the WRS-3 cycle, and thus the 18 days of the DRC-18. Total DRC-18 sun-lit daily scene counts range from a low of 762 to a high of 887, driven primarily by this geographic variation in coverage. The longest interval present in the DRC-18 is 84 scenes, occurring on days 8, 9, 13 and 16 of the DRC-18, spanning Eurasia and Africa. This may not be the longest interval possible, but it is the longest interval in this reference case.

3.4.2.1 LandIS Earth Observation Imaging Processes

The LandIS instrument onboard the L10 Observatory is a nadir viewing, cross-track observing, electro-optical imaging sensor. The Opto-Mechanical Assembly (OMA) contains all of the optical and mechanical assemblies required to collect Earth and calibration data including scanning optics, focal planes, and calibration sources. LandIS views the scene using a scanning mirror that scans the field of view such that each ~7.5 seconds LandIS observes a portion of the Earth, plus two calibration sources (Blackbody and Space View) before returning to its original starting position.

The LandIS focal planes are scanning arrays. Eight arrays are set up in a staggered configuration on the Visible/Shortwave (VIS/SWIR) Focal Plane Assemblies (FPAs) and four on the TIR FPA. As the arrays are scanned across the desired swath a scan's worth of raw data is built up in the following way:

1. For a given sample in time, the array is collapsed down into a line scanner by a process called Time Delay Integration. Time Delay Integration combines the output of several detectors in the array line to increase Signal-to-Noise Ratio (SNR) and occurs onboard the Sensor Chip Assembly (SCA) chip. The effect produces one "detector" per time sample.
2. The effective line of detectors is scanned over time across the desired swath. This produces multiple samples for each detector.
3. Onboard processing then spatially aggregates these detectors to improve performance and make a larger sample on the ground.
4. The collection of these aggregated samples makes up the scan's worth of data sent to the compression engine for packaging into the instrument data blocks and transmitted to the spacecraft.

In order to maintain the same number of these samples across the swath as the sensor traverses the orbit and latitude changes, scan rate of the mirror and frame rate of the focal planes will change up to 40 times per orbit.

With each scan being approximately 7.5 seconds in duration, it takes 4 full scans to completely cover a single WRS-3 scene because each scan covers 52km in-track and each scene is 168km.

3.4.3 Night Imaging (Collection Type-6)

Night Imaging consists of nadir-viewing Earth imaging collections, taken when there is no direct solar illumination of the ground or when the sun angle is less than the sun-lit land threshold of 5 degrees. The DRC-18 night targets consist of large areas being monitored for fires, active volcanoes, and spurious urban areas and occur throughout the DRC-18.

3.4.4 Calibrations

Accurate radiometry is a hallmark of the Landsat program. Maintaining the required accuracy requires several routine calibration collections during a cycle. The full set of on-orbit LandIS calibration activities may be found within the LandIS Instrument Operations Concept Document (LandIS-SE14-001). As the frequency of the calibration activities vary greatly, only a sub-set of these calibrations is represented within the DRC-18, with the focus being on calibrations that occur during a nominal WRS-3 18-day cycle and those that may require some spacecraft slews to achieve the proper viewing angles as noted within the following descriptions.

3.4.4.1 Blackbody and Space View Calibrations During Earth Viewing Observations

During each ~7.5-second scan period LandIS views one full-aperture onboard calibration source and deep space in addition to the Earth. These frequent calibration sources include a Blackbody with accurately characterized emissivity and temperature,

and view of space to act as a known low-radiance source to provide information for radiometric calibration. For the purposes of the DRC-18, these collections are part of Collection Type-1 “Nominal Earth Imaging Acquisition”.

It should be noted that the Blackbody and Space View calibration sources are also used during other types of calibrations, such as lunar and lamp calibrations, and are included as part of these calibration time periods.

3.4.4.2 LandIS Stim Lamp Calibration (Collection Type-2)

Every day, every 2 weeks, and every 3-6 months the scan mirror will swing around to view the working, bimonthly, and pristine stim lamps respectively to provide relative radiometric calibration response trending between solar calibrations. This activity would take around 4 to 5 minutes and can be scheduled during the nighttime orbital node. The scan mirror will dwell on the Space View port for one minute before and after the lamp collections. The scan mirror will dwell on the stim lamps for up to two minutes. During the Space View and lamp collections, data will be collected for 1 out of every 6 seconds resulting in 10 collections of Space View data before and after the 20 collections of lamp data. The data collected will be unaggregated detector data and will not be compressed. Lamp calibrations occur on Paths 176, 181, 168, 173, 160, 183, 170, 175, 180, 185, 203, 172, 177, 182, 169, 174, 179, 184 and 171.

3.4.4.3 LandIS Solar Diffuser Calibration (Collection Type-3)

Every 6 days, the spacecraft will perform a yaw maneuver, and the scan mirror will swing around to view the solar diffuser to provide reflective gain corrections and measures of uniformity. The solar diffuser is positioned equal and opposite nadir to get the same angle of incidence on the scan mirror during collection. This activity must be conducted on the dark side of the Earth’s terminator crossing at the South Pole. This allows for 57 seconds of full solar illumination. The scan mirror will dwell for 2 minutes on the solar diffuser to ensure capture of the full 57 seconds of illumination. On either side of the solar diffuser collect, a Space View dwell should be collected for offset. Both the solar diffuser collections and Space View collections will be taken for 1 out of every 6 seconds and the data will be unaggregated and uncompressed. At the conclusion of the collect, the spacecraft can yaw back to nominal orientation. The magnitude of the needed yaw maneuver (roughly between 2 and 24 degrees) will change throughout the year. There are 3 Solar (Diffuser) Calibrations within the DRC-18, and they appear on Day 3 Path 204 starting at Row 124, Day 9 Path 234 starting at Row 122, and Day 15 Path 174 starting at Row 120.

3.4.4.4 LandIS Lunar Calibration (Collection Type-5)

Every 30 days spacecraft will perform a roll maneuver and scan mirror will scan in only the Earth View to view the full disk of the moon at a constant phase. This provides a source of truth for solar diffuser reflectance updates, stray light monitoring, ghosting, edge response, and band-to-band registration. The roll would occur on the nighttime side of a single orbit and may also require some small pitch maneuvers by the spacecraft to help center the disk. A special scan and view profile will be created to collect 3 degrees in 2.25 sec period without Space View or Blackbody views. This

covers the moon plus views of space on either side of moon. A total of 10 swaths will be collected as the moon traverses the SCAs and detectors within LandIS. After the collections are complete, the spacecraft will return to nominal orientation in time to begin nominal Earth View observations. The Lunar calibrations are scheduled on Day 5 of the DRC-18 on Path 178 starting on Row 190.

In addition to the above lunar calibration, but not represented within this DRC, every 3-6 months the spacecraft will perform the roll maneuver for 2 orbits instead of the usual 1. During the second orbit, the spacecraft will perform smaller roll maneuvers to move the angle of incidence the scan mirror views the disk. This provides a secondary response vs. scan measurement.

3.4.5 Off-Nadir Observations (Collection Type-7)

Off-nadir scene acquisitions are special observations that require the spacecraft to slew to orientation that allows for acquisitions that are not on the current WRS-3 path swath. These off-nadir acquisitions are typically done to acquire scenes one path east or west of the current path; however, it should be noted that in some rare cases slewing to a orientation that is not a full path east or west of the current path may also occur. These observations may be done during the day-lit portion of the orbit (descending), or on the night side (ascending). Examples of off-nadir imaging for full path offsets from the current path on both the ascending and descending sides of the orbit are contained within the DRC-18.

The ascending orbit off-nadir acquisition is performed off the coast of California on Day 9. On Path 162, the off-nadir image collection is arranged to be one path east of the ground track, such that path 161, rows 211, 212, and 213 will be imaged.

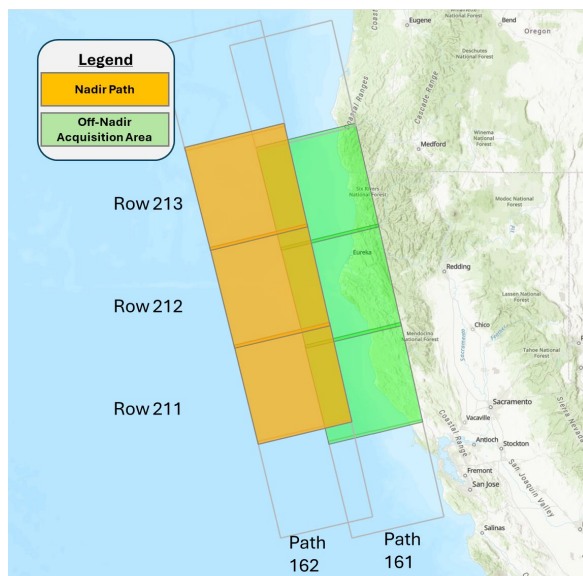


Figure 3-2. Ascending Off-Nadir Acquisition of Path 161 While On Path 162

The descending orbit off-nadir acquisitions are performed off the coast of Australia on Day 2 and Day 16. On Day 2 during Path 145, an off-nadir collect is set for row 84 in order to acquire Path 144 Row 84. On Day 16 during Path 143, an off-nadir collect is set for Row 84 in order to acquire Path 144 Row 84 again.

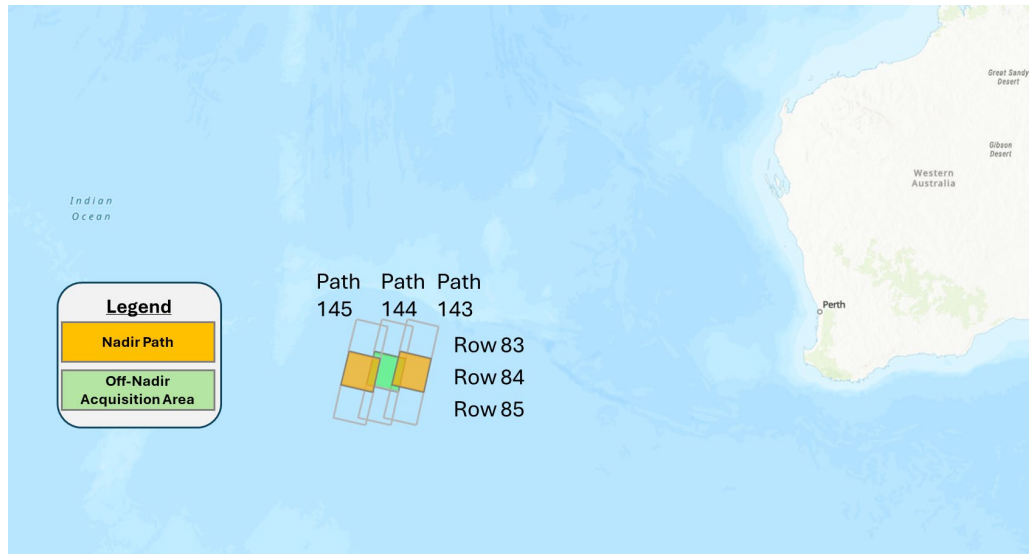


Figure 3-3. Descending Off-Nadir Acquisitions of Path 144 While On Path 143 and 145

Section 4 Single Observatory, DRC-18 Figures

This section provides a graphical representation of the data acquisitions defined within the DRC-18 Microsoft Excel spreadsheet for a single observatory.

Orbit # per day	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Mar 12, 2022	122	140	158	176	194	212	230	248	1	19	37	55	73	91	109
Mar 13, 2022	127	145	163	181	199	217	235	253	6	24	42	60	78	96	114
Mar 14, 2022	132	150	168	186	204	222	240	258	11	29	47	65	83	101	
Mar 15, 2022	119	137	155	173	191	209	227	245	263	16	34	52	70	88	106
Mar 16, 2022	124	142	160	178	196	214	232	250	3	21	39	57	75	93	111
Mar 17, 2022	129	147	165	183	201	219	237	255	8	26	44	62	80	98	
Mar 18, 2022	116	134	152	170	188	206	224	242	260	13	31	49	67	85	103
Mar 19, 2022	121	139	157	175	193	211	229	247	265	18	36	54	72	90	108
Mar 20, 2022	126	144	162	180	198	216	234	252	5	23	41	59	77	95	113
Mar 21, 2022	131	149	167	185	203	221	239	257	10	28	46	64	82	100	
Mar 22, 2022	118	136	154	172	190	208	226	244	262	15	33	51	69	87	105
Mar 23, 2022	123	141	159	177	195	213	231	249	2	20	38	56	74	92	110
Mar 24, 2022	128	146	164	182	200	218	236	254	7	25	43	61	79	97	
Mar 25, 2022	115	133	151	169	187	205	223	241	259	12	30	48	66	84	102
Mar 26, 2022	120	138	156	174	192	210	228	246	264	17	35	53	71	89	107
Mar 27, 2022	125	143	161	179	197	215	233	251	4	22	40	58	76	94	112
Mar 28, 2022	130	148	166	184	202	220	238	256	9	27	45	63	81	99	
Mar 29, 2022	117	135	153	171	189	207	225	243	261	14	32	50	68	86	104
Mar 30, 2022	122	140	158	176	194	212	230	248	1	19	37	55	73	91	109
Mar 31, 2022	127	145	163	181	199	217	235	253	6	24	42	60	78	96	114

Figure 4-1. DRC-18 Path Sequence per Day

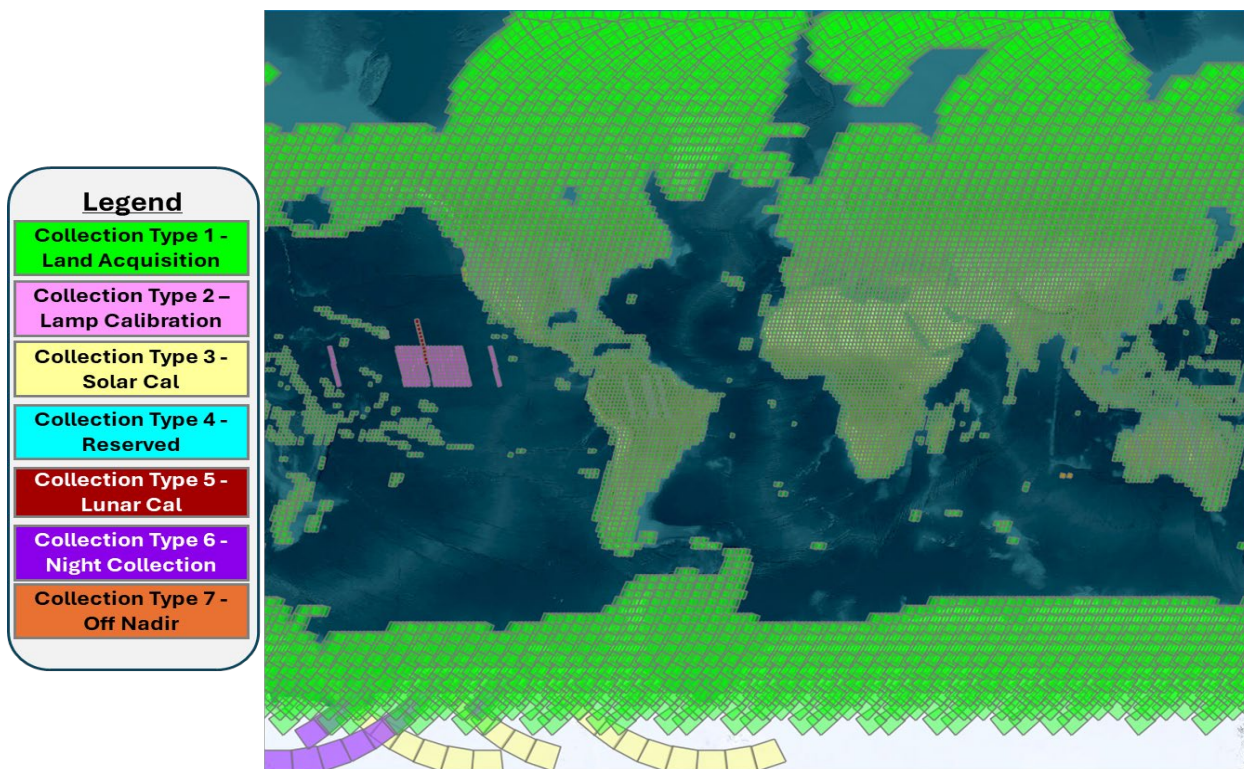


Figure 4-2. DRC-18 All Acquisitions

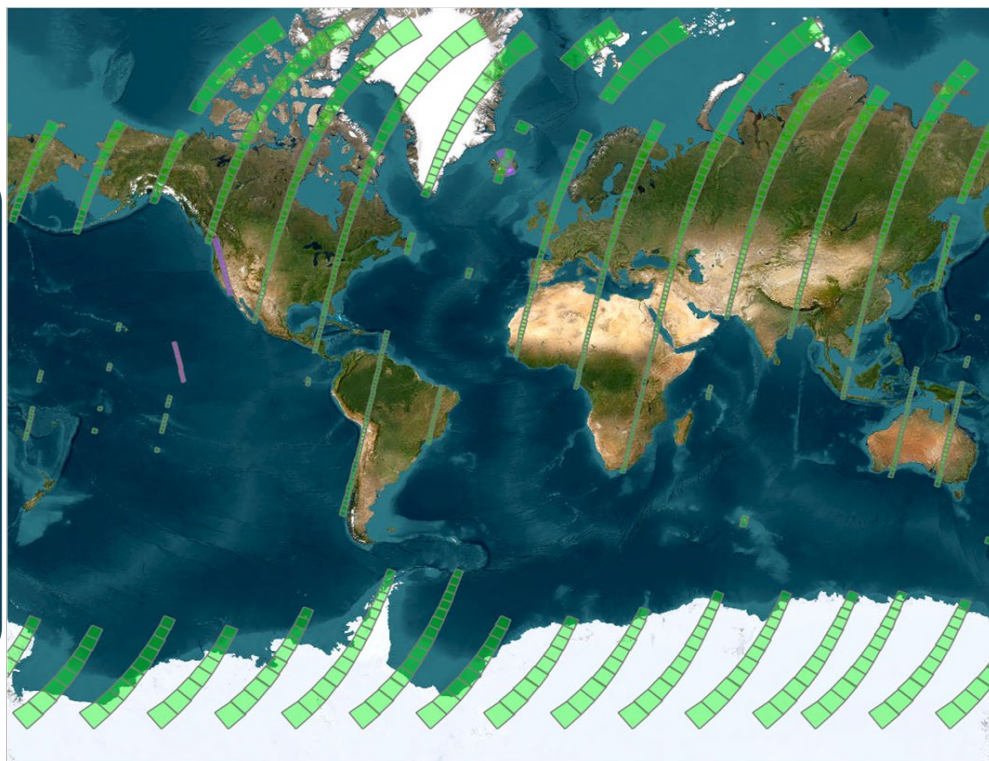


Figure 4-3. DRC-18 Day 1 All Acquisitions

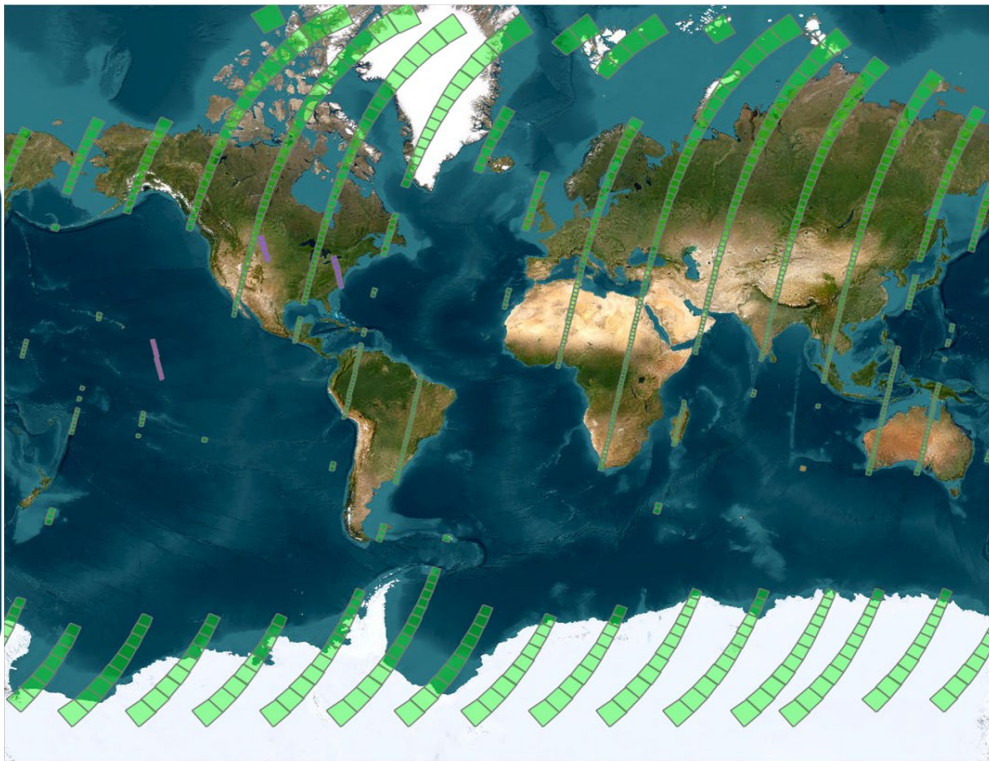


Figure 4-4. DRC-18 Day 2 All Acquisitions

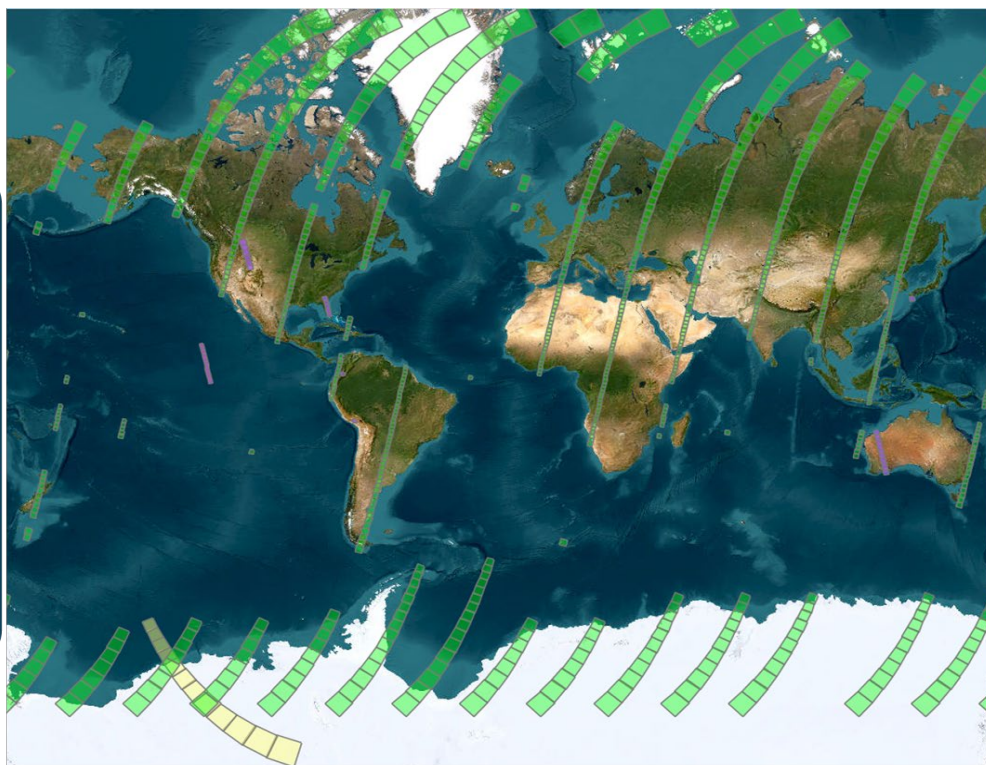


Figure 4-5. DRC-18 Day 3 All Acquisitions

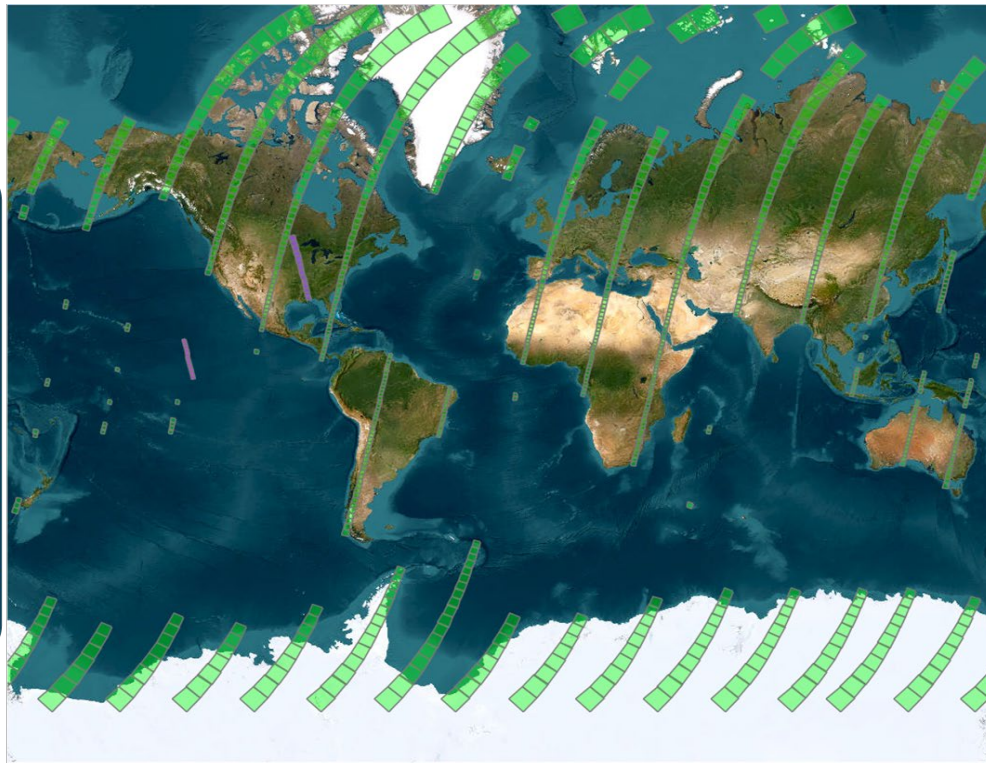


Figure 4-6. DRC-18 Day 4 All Acquisitions

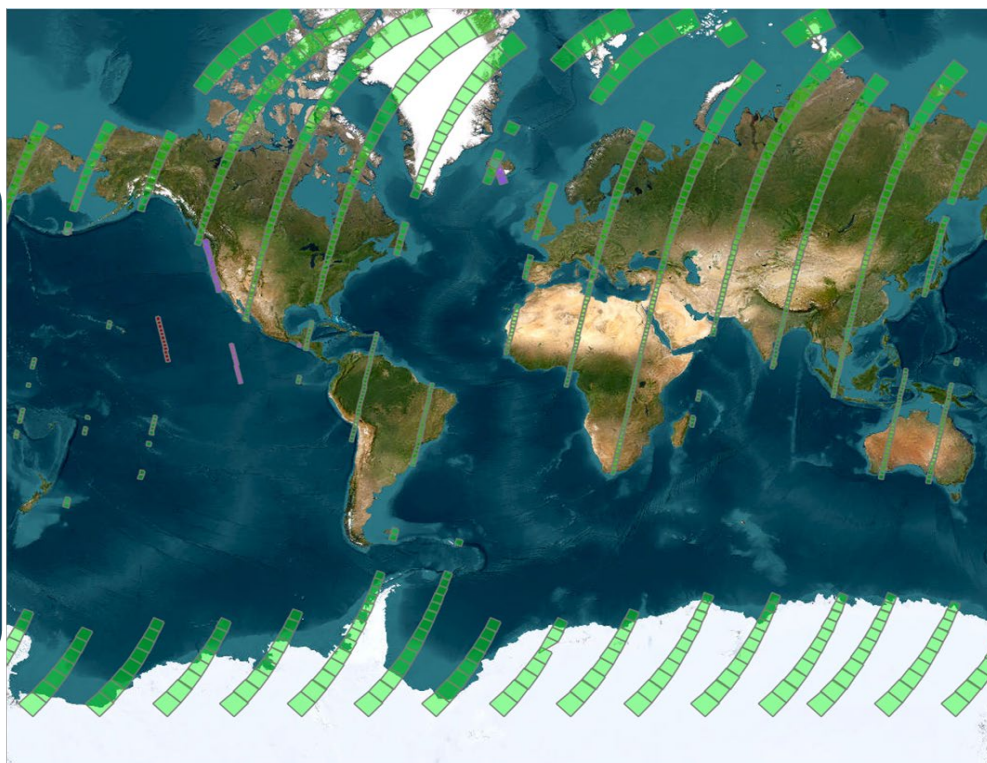


Figure 4-7. DRC-18 Day 5 All Acquisitions

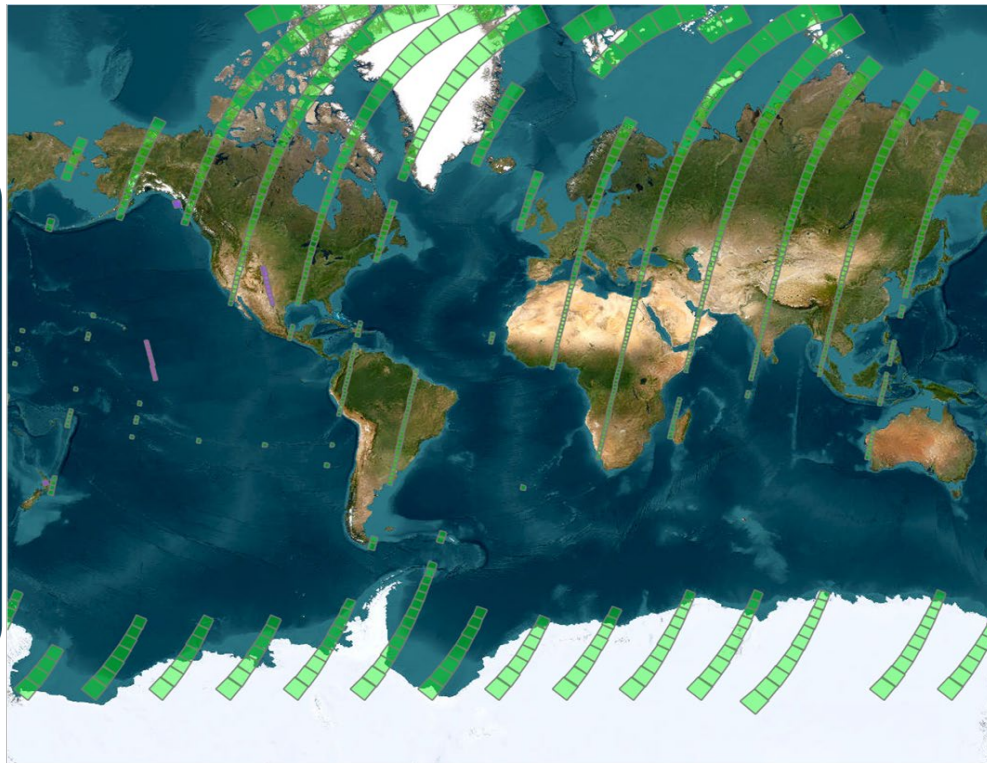


Figure 4-8. DRC-18 Day 6 All Acquisitions

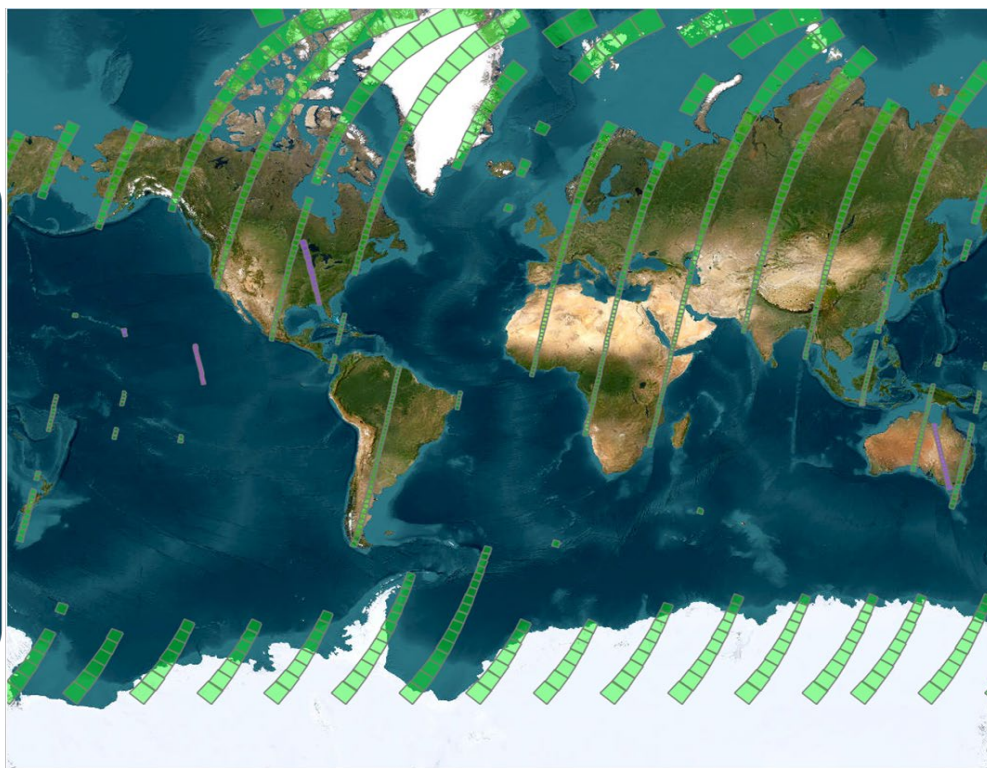


Figure 4-9. DRC-18 Day 7 All Acquisitions

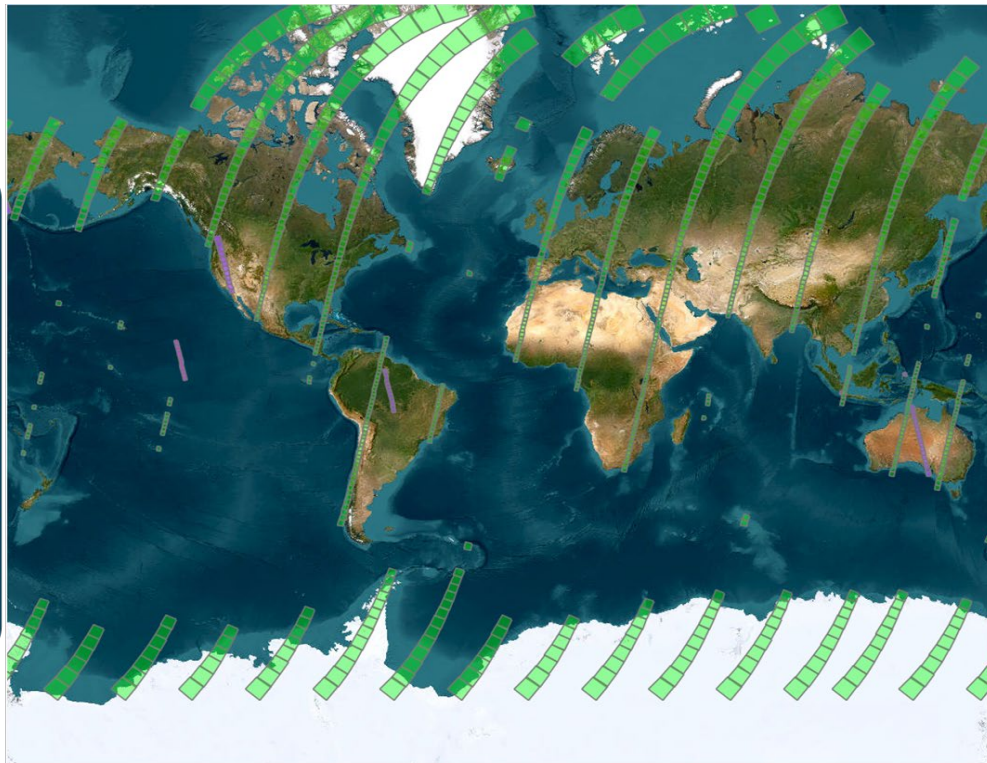


Figure 4-10. DRC-18 Day 8 All Acquisitions

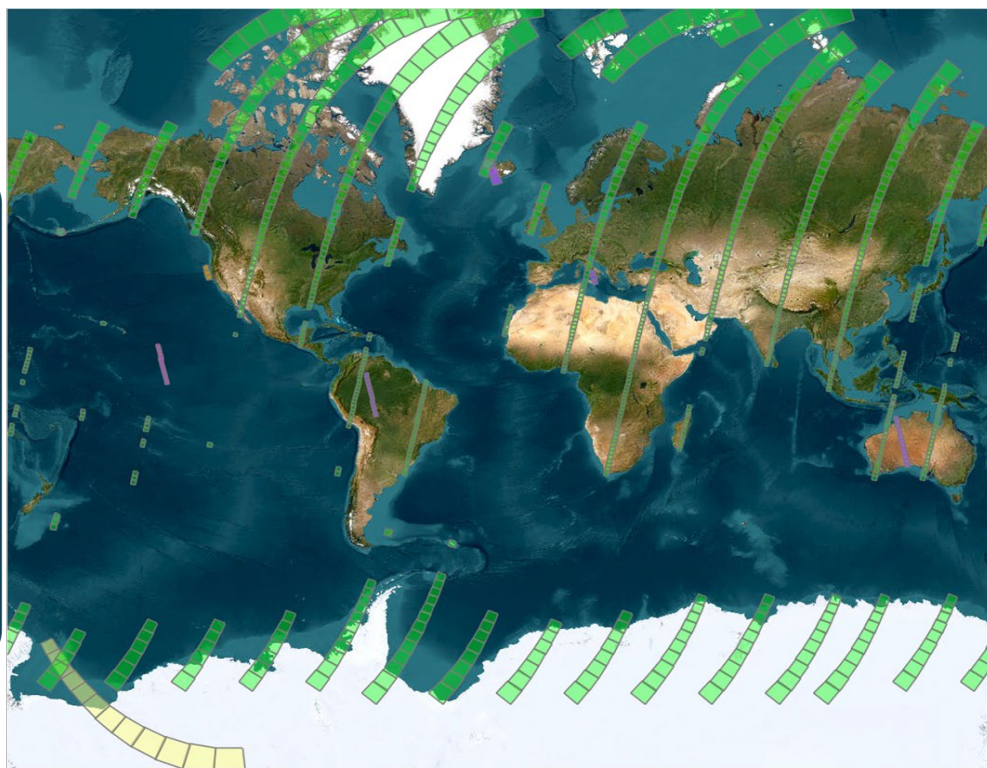


Figure 4-11. DRC-18 Day 9 All Acquisitions

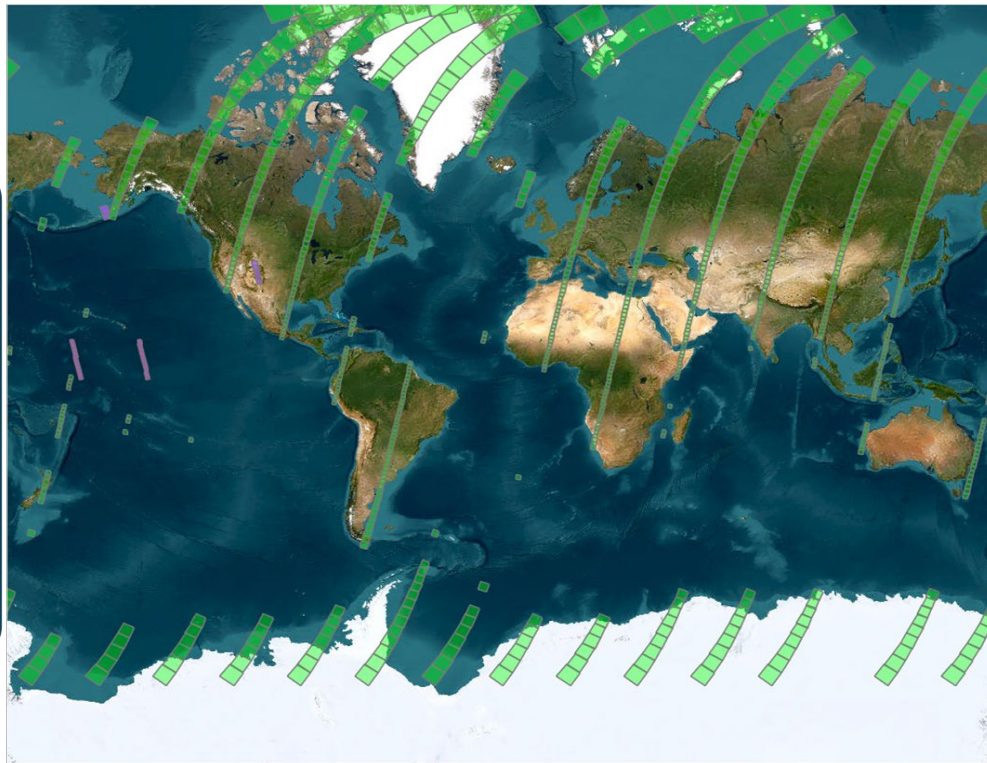


Figure 4-12. DRC-18 Day 10 All Acquisitions

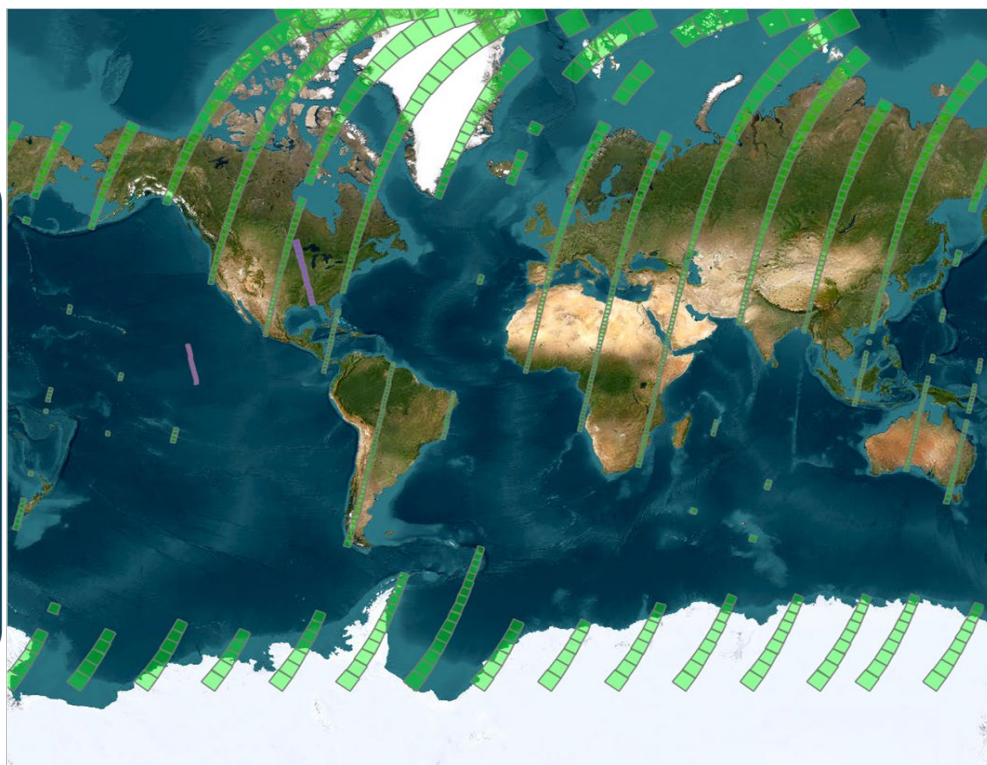


Figure 4-13. DRC-18 Day 11 All Acquisitions

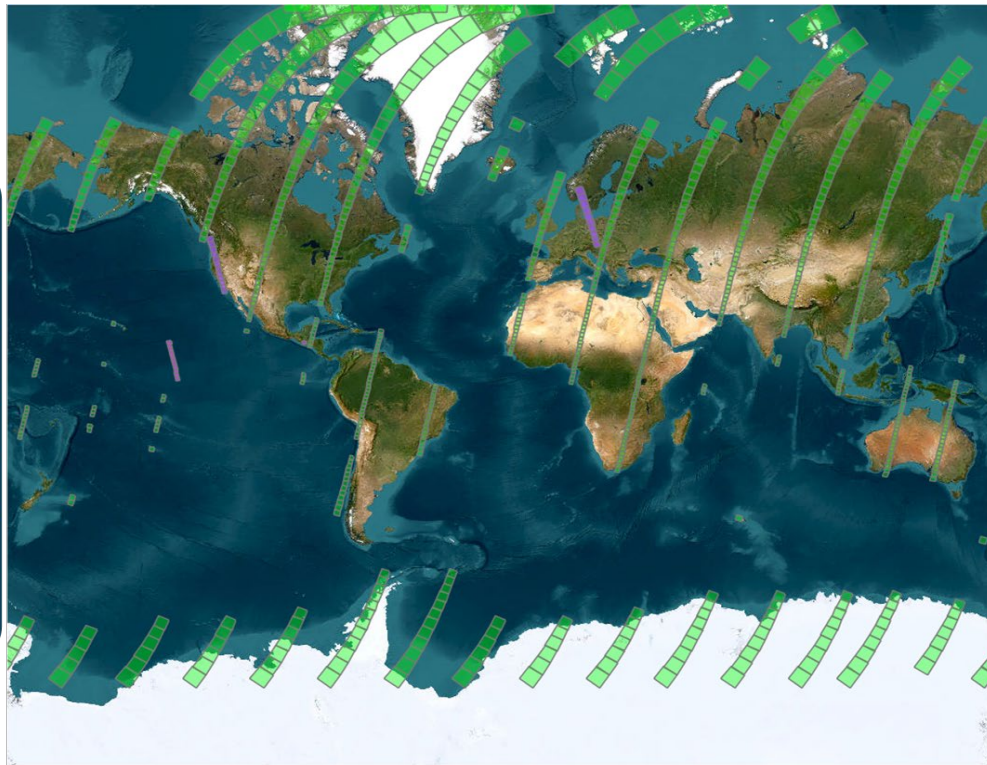


Figure 4-14. DRC-18 Day 12 All Acquisitions

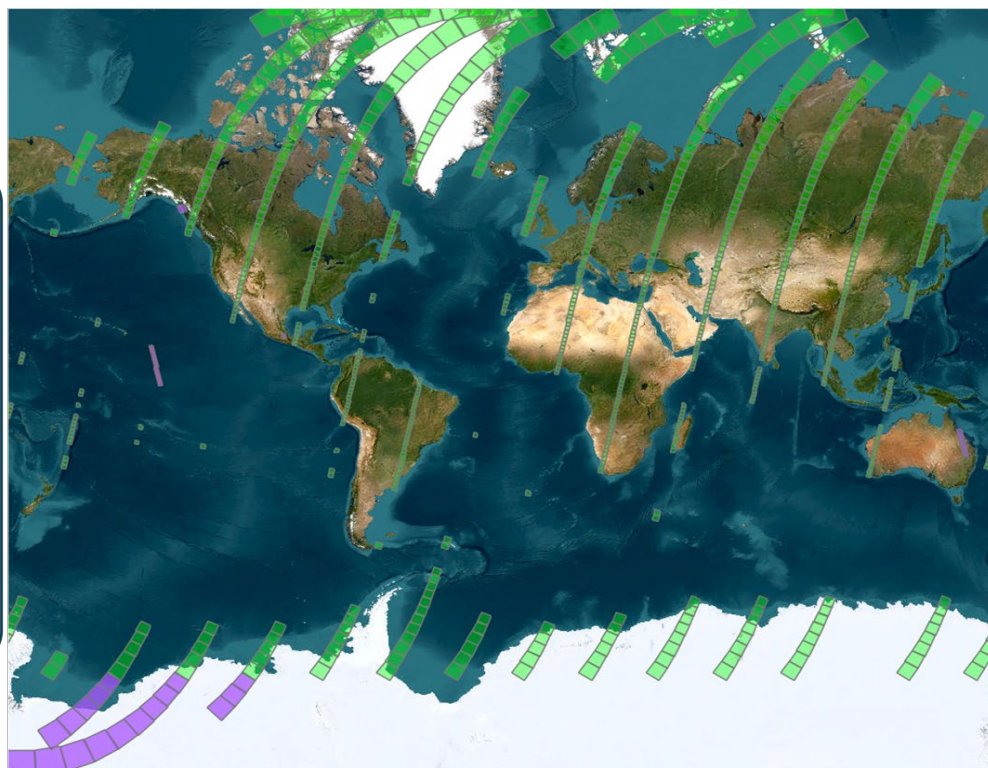


Figure 4-15. DRC-18 Day 13 All Acquisitions

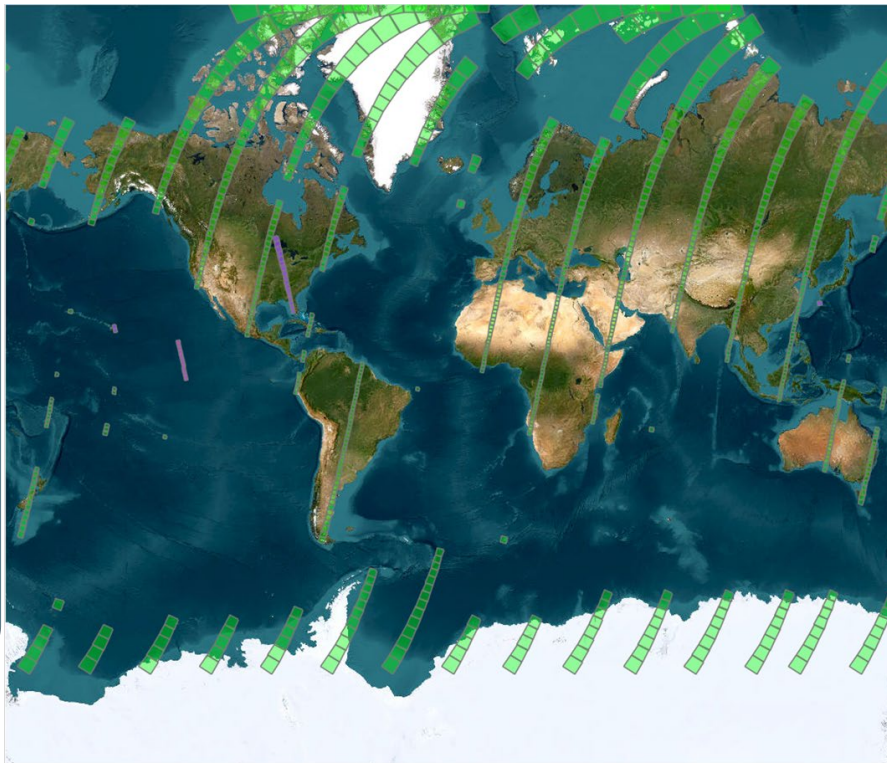


Figure 4-16. DRC-18 Day 14 All Acquisitions

Legend	
Collection Type 1 -	Land Acquisition
Collection Type 2 -	Lamp Calibration
Collection Type 3 -	Solar Cal
Collection Type 4 -	Reserved
Collection Type 5 -	Lunar Cal
Collection Type 6 -	Night Collection
Collection Type 7 -	Off Nadir

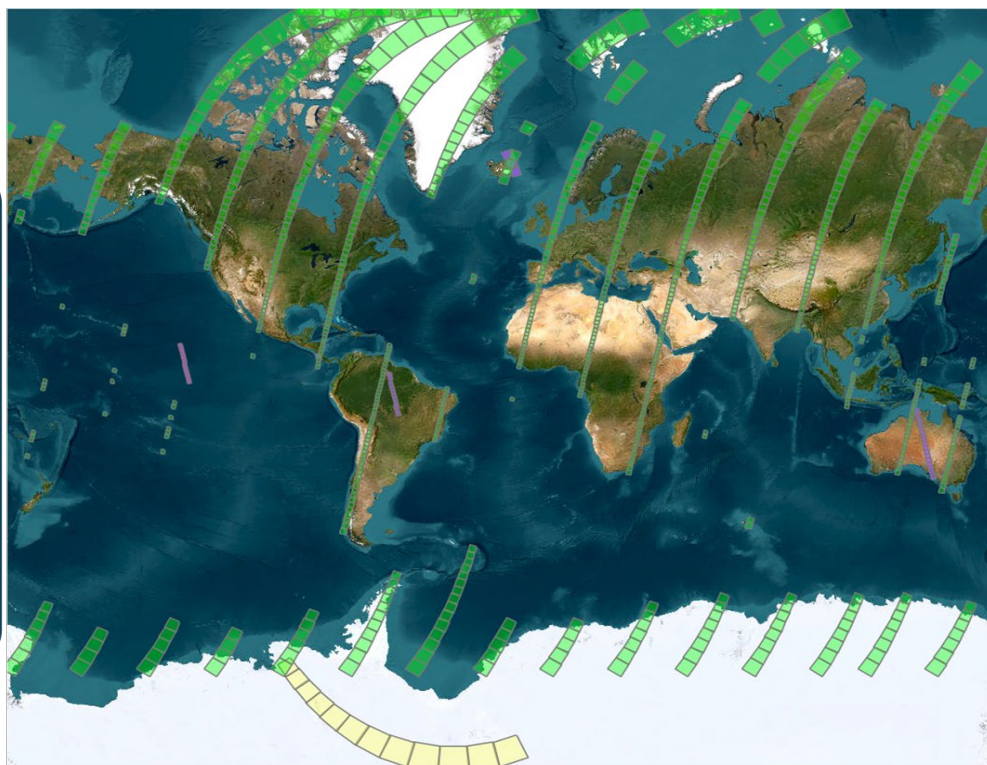


Figure 4-17. DRC-18 Day 15 All Acquisitions

Legend	
Collection Type 1 -	Land Acquisition
Collection Type 2 -	Lamp Calibration
Collection Type 3 -	Solar Cal
Collection Type 4 -	Reserved
Collection Type 5 -	Lunar Cal
Collection Type 6 -	Night Collection
Collection Type 7 -	Off Nadir

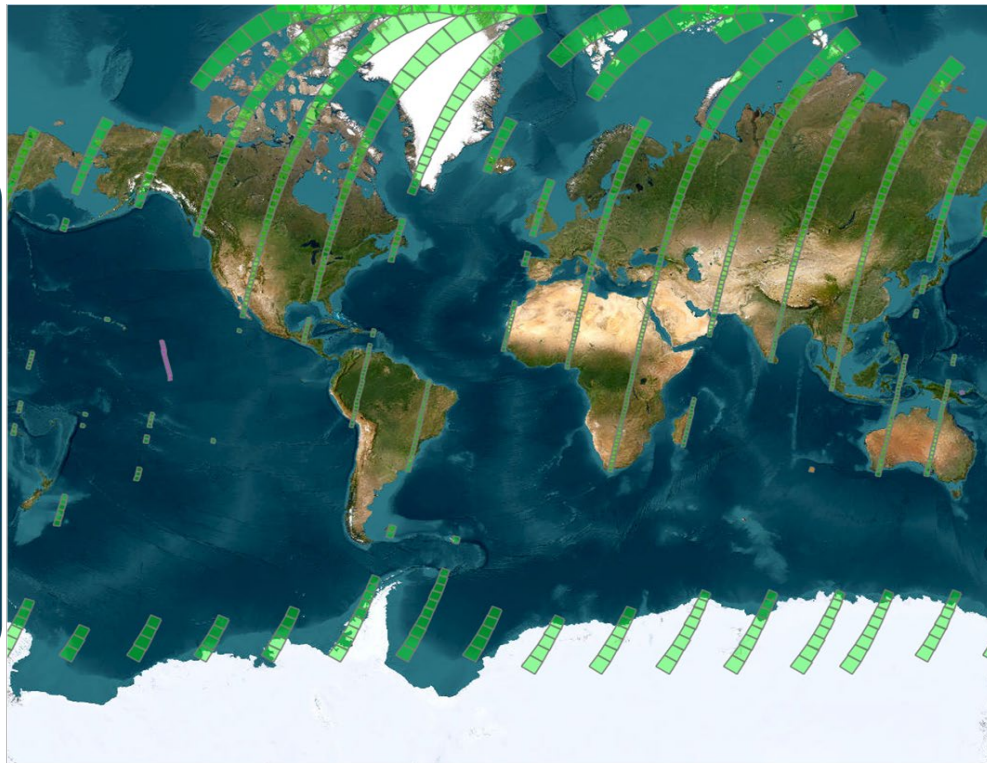


Figure 4-18. DRC-18 Day 16 All Acquisitions

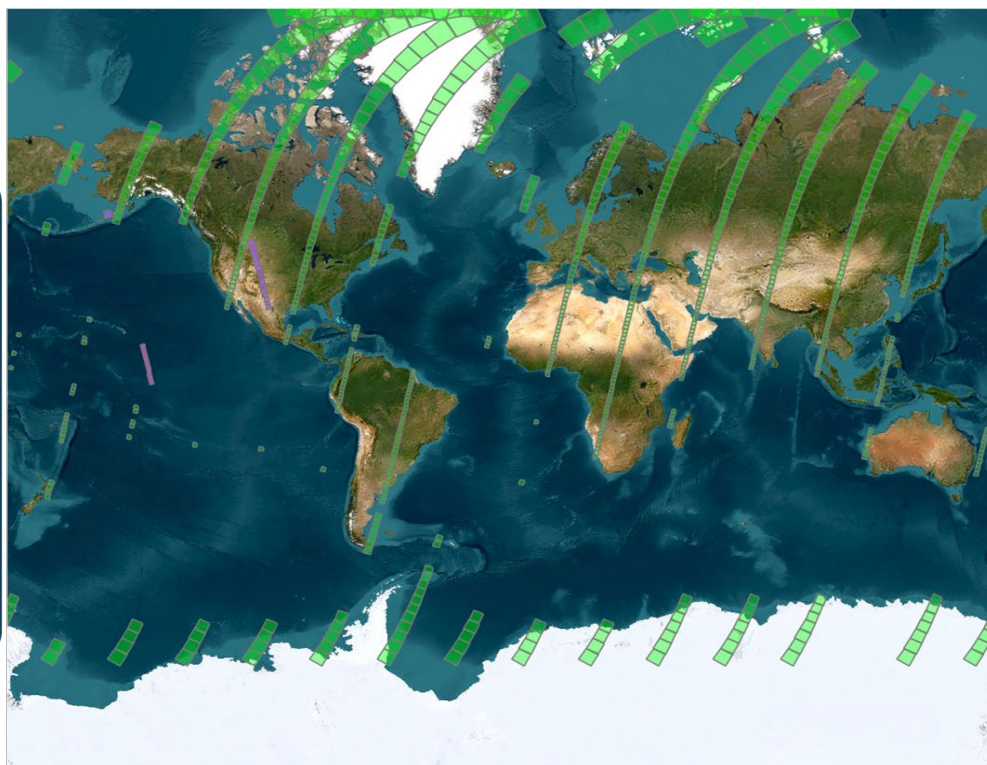


Figure 4-19. DRC-18 Day 17 All Acquisitions

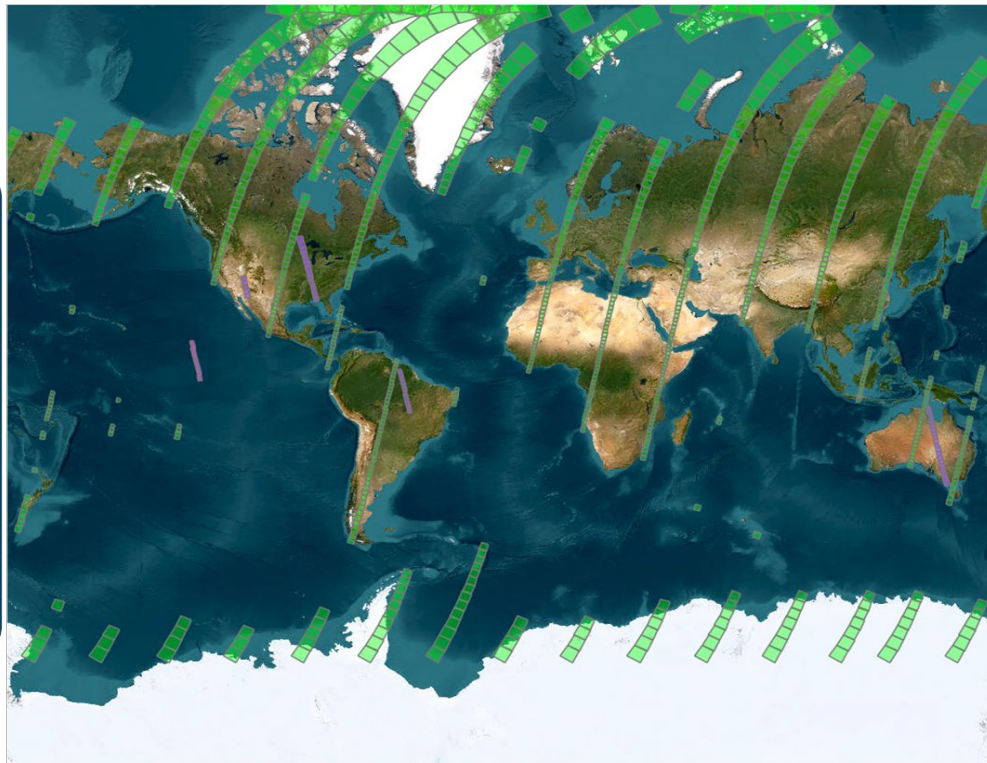


Figure 4-20. DRC-18 Day 18 All Acquisitions

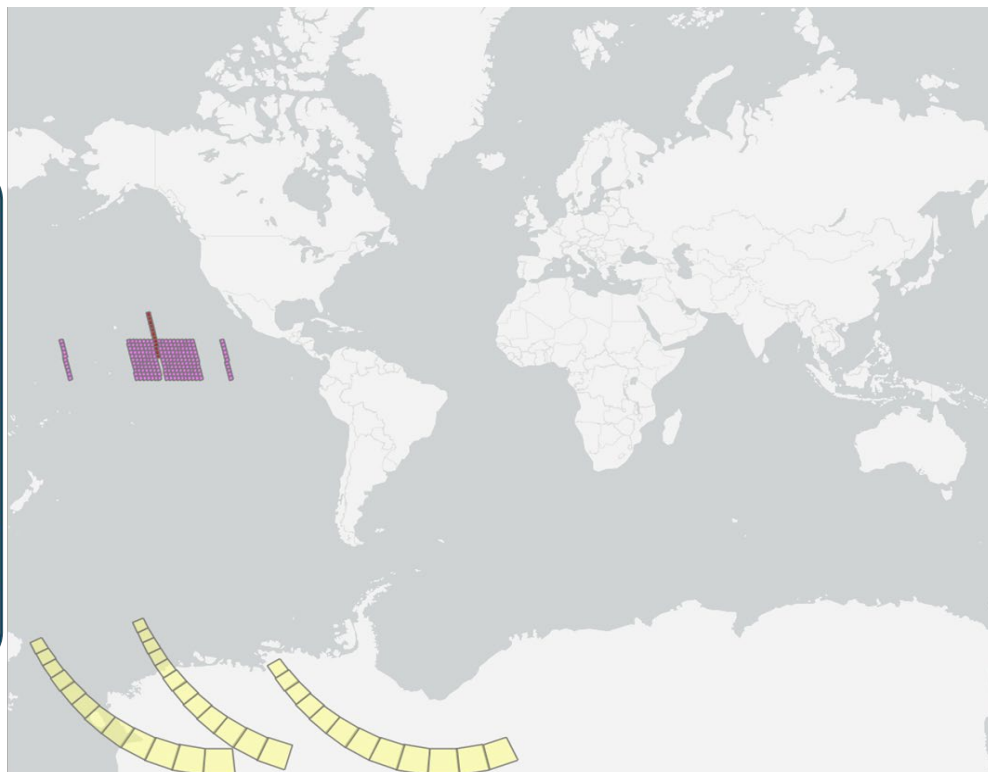


Figure 4-21. DRC-18 All Instrument Calibrations



Figure 4-22. DRC-18 Off-Nadir Acquisitions

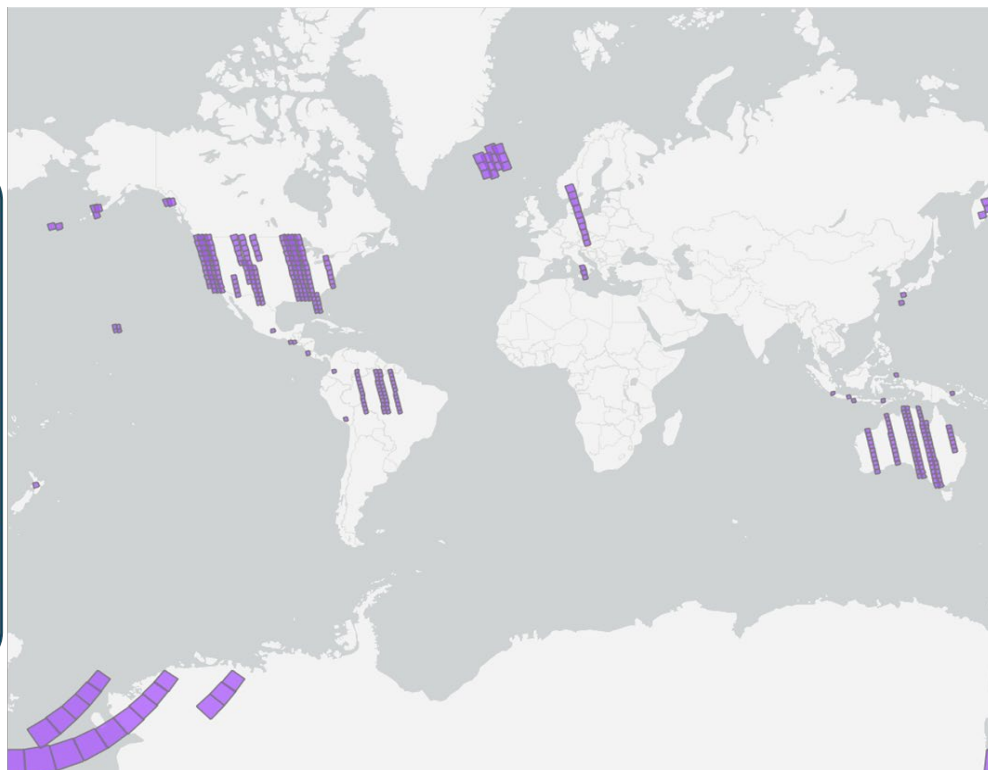


Figure 4-23. DRC-18 Night Acquisitions

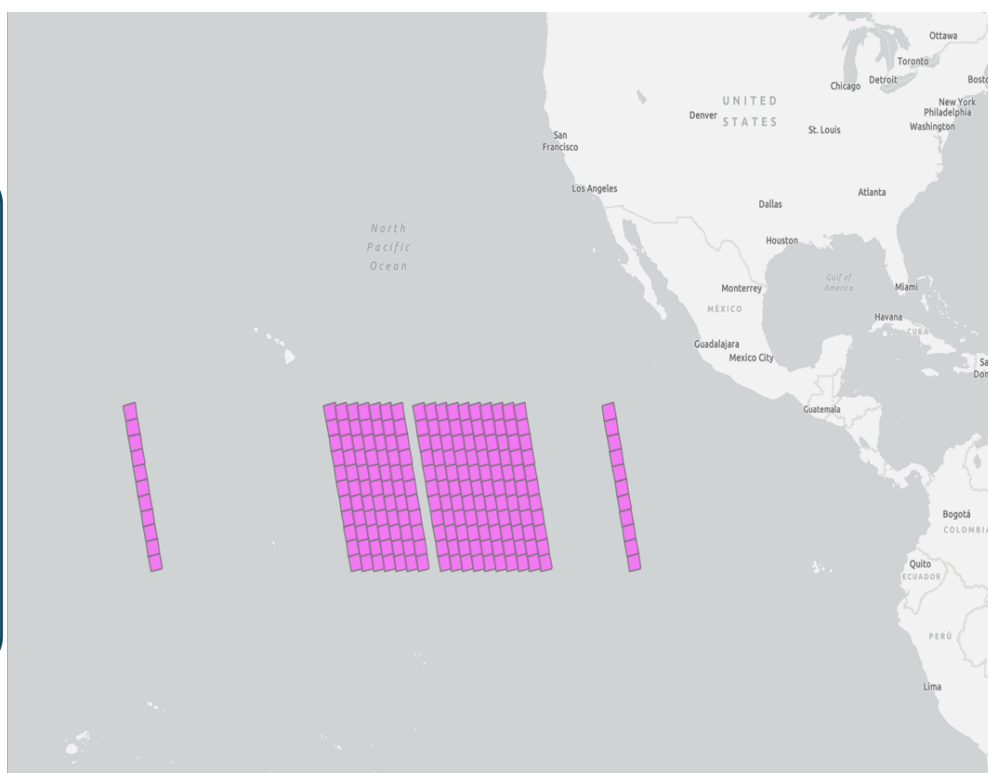
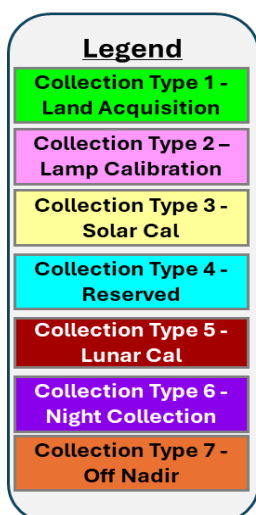


Figure 4-24. DRC-18 Lamp Calibrations

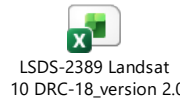
Appendix A Acronyms

Cal/Val	Calibration and Validation
CCB	Configuration Control Board
CI	Configuration Item
CM	Configuration Management
CR	Change Request
DCN	Document Control Number
DOI	Department of the Interior
DRC	Design Reference Case
EROS	Earth Resources Observation and Science
ERTS	Earth Resources Technology Satellite
FOT	Flight Operations Team
FPA	Focal Plane Assembly
GSD	Ground Sample Distance
GSFC	Goddard Space Flight Center
L8	Landsat 8
L9	Landsat 9
L10	Landsat 10
LandIS	Landsat Instrument Suite
LCR	Land Collection Request
NASA	National Aeronautics and Space Administration
NSLRSDA	National Satellite Land Remote Sensing Data Archive
OMA	Opto-Mechanical Assembly
PLRA	Program-Level Requirements Appendix
SCA	Sensor Chip Assembly
SLI	Sustainable Land Imaging
SMRD	Science Mission Requirements Document
SNR	Signal-to-Noise Ratio
TIR	Thermal Infrared
USGS	U.S. Geological Survey
UTC	Coordinated Universal Time
VSWIR	Visible to Shortwave-Infrared
WRS	Worldwide Reference System

Appendix B DRC-18 Table

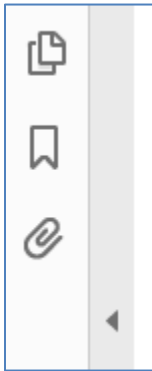
The DRC-18 table is attached to this document.

If viewing this document in Word, double-click the icon to open the table.



If viewing this document in a PDF, to open the DRC-18 table:

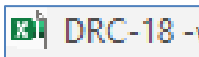
1. Ensure the PDF is open in the Adobe app, not in a browser
2. Click the arrow to expand the sidebar, if needed



3. Click the paper clip icon



4. Double-click the attachment



Alternatively, a copy of the table can be found at

<https://edcdocushare.cr.usgs.gov/docushare/dsweb/View/Collection-46890>